

VISHWAA SHAH

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EDUCATION

M.S. Computer Science - University of Texas at Dallas

Expected May 2027

Relevant: Machine Learning, Network Security, Database Design, Statistical Methods for Data Science

B.Tech, Computer Engineering - Sarvajani College of Engineering & Technology, India

June 2025

TECHNICAL SKILLS

AI & Agents	Agentic AI (Claude Code, Claude Cowork), LangChain, RAG, pgvector, Chatbot Dev, Tool/Function Calling, Scikit-learn, Isolation Forest, Deep Learning
Backend	Python, Node.js, NestJS, Flask, REST API Design, PHP, WordPress (themes, plugins, REST API), Java, TypeScript, DSA
Frontend	React.js, HTML5, CSS3, Power BI
Cloud & Databases	AWS (Lambda, S3), Azure App Service, Docker, Kubernetes, GitHub Actions (CI/CD), Git, Postman, Linux, PostgreSQL (pgvector), MySQL
Security	RSA, Diffie-Hellman, HMAC-SHA256, OpenSSL, HIPAA Awareness, Privacy-First Design

PROFESSIONAL EXPERIENCE

Software Developer Intern | Palm Infotech, Surat, India

Jan 2025 – Apr 2025

- Built **10+ REST APIs** using **NestJS** and **MySQL** for **real-time data processing**; improved system **efficiency ~30%** via API response and **query optimization**.
- Integrated third-party mapping APIs enabling real-time GPS tracking with **<250 ms latency** under **~300 concurrent users**.
- Led end-to-end **Agile SDLC** in a team of 4; achieved **85%+ API test coverage** via Postman/Swagger, cutting regression issues significantly.

Computer Science Grader | University of Texas at Dallas

Sep 2025 – Present

- Mentored **80+ students** in Data Structures & Algorithms; evaluated **300+ submissions/semester** across trees, graphs, sorting, and Big-O complexity.
- Conducted 15+ one-on-one debugging sessions per semester across 2 evaluation pipelines, directly boosting assignment scores for 10–20 students through targeted algorithm design feedback.

PROJECTS

Trackly – AI-Native Project & Issue Tracking Platform | Personal | UT Dallas

Mar 2026

Stack: React.js, Node.js, PostgreSQL, RAG (LangChain + pgvector), Docker, GitHub Actions

- Engineered full-stack issue tracking platform (React.js + Node.js + PostgreSQL) with **15+ REST API** endpoints supporting complete ticket lifecycle across **3 project workspaces**.
- Built **RAG pipeline** (LangChain + pgvector) powering AI chatbot with semantic search across **500+ indexed issues**, delivering context-aware resolution hints in **<2s via tool/function calling**.
- Reduced duplicate ticket creation by **~40% through auto-suggestion** of similar past issues; deployed via Docker + GitHub Actions CI/CD with zero-downtime releases.

AI-Powered Security Log Anomaly Detector | Personal | UT Dallas

Apr 2026

Stack: Python, Flask, Scikit-learn, Isolation Forest, HTML5/CSS3, Azure App Service, GitHub Actions

- Deployed Isolation Forest anomaly detection model processing 10K+ log entries/day on Azure App Service, achieving **92% precision** in identifying brute-force and request-spike attack patterns.
- Architected end-to-end ML pipeline (**ingestion → inference → REST API → dashboard**) with **<3s detection latency** and **99%+ uptime** over a 30-day production window.

Secure Relay-Based Chat System | Personal | UT Dallas

Oct 2025

Stack: Python (sockets), RSA, Ephemeral Diffie-Hellman, HMAC-SHA256, OpenSSL

- Implemented ephemeral Diffie-Hellman key exchange with **per-session forward secrecy**, completing **cryptographic handshake** in **<50ms across 20+ concurrent clients**.
- Secured message integrity using **HMAC-SHA256 + nonce validation**, recording 0 successful replay or tampering attacks across **100+ adversarial test sessions**.